



CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

MATHEMATICS

UNIT 2 – PAPER 02

2½ hours

06 JUNE 2000 (a.m.)

This examination paper consists of **THREE** sections: Module 2.1, Module 2.2, and Module 2.3.

Each section consists of 2 questions.

The maximum mark for each section is 50.

The maximum mark for this examination is 150.

This examination paper consists of 4 pages.

INSTRUCTIONS TO CANDIDATES

1. **DO NOT** open this examination paper until instructed to do so.
2. Answer **ALL** questions from the **THREE** sections.
3. Unless otherwise stated in the question, all numerical answers **MUST** be given exactly **OR** to three significant figures as appropriate.

Examination materials

Mathematical formulae and tables

Electronic calculator

Ruler and compass

SECTION 1 (MODULE 2.1)

Answer BOTH questions.

1. (a) Given that the derivative at x of the exponential function e^x is e^x , show that

(i) by using $e^{-x} = \frac{1}{e^x}$, the derivative of e^{-x} is $-e^{-x}$ [4 marks]

(ii) if $u = e^x + e^{-x}$, $v = e^x - e^{-x}$, then

$$\frac{du}{dx} = v \text{ and } \frac{dv}{dx} = u$$
 [6 marks]

(iii) $\frac{d^2u}{dx^2} = u$. [3 marks]

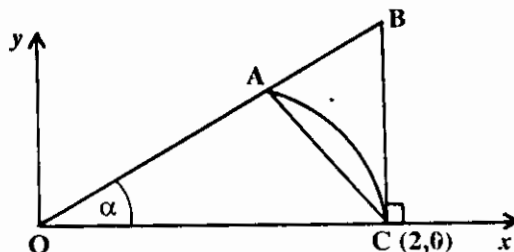
- (b) Evaluate the following integrals:

(i) $\int x^3 e^x dx$ [6 marks]

(ii) $\int x^3 \ln 4x dx, x > 0$ [6 marks]

Total 25 marks

2. The origin, O, of the xy - plane is the centre of a circle of radius 2 units in the plane. C is the point with coordinates (2,0) and A is a point on the circle in the first quadrant so that arc, CA, subtends an acute angle, α radians, at O. B is the point on OA produced so that CB is perpendicular to the x -axis.



- (a) Calculate, in terms of α , the areas of

(i) the triangle OAC [3 marks]

(ii) the sector OAC, and [3 marks]

(iii) the triangle OBC. [4 marks]

- (b) By considering the areas of the triangle OAC, sector OAC and triangle OBC, show that

$$\cos \alpha < \frac{\sin \alpha}{\alpha} < 1. \quad [6 \text{ marks}]$$

- (c) Hence, show that $\frac{\sin \alpha}{\alpha} \rightarrow 1$ as $\alpha \rightarrow 0$. [3 marks]

- (d) By using the substitution $\alpha = 2x$, find $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x}$. [6 marks]

Total 25 marks

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SECTION 2 (MODULE 2.2)

Answer BOTH questions.

3. (a) (i) Given that S_r denotes the sum of the first r positive integers, prove by mathematical induction, that

$$S_r = \frac{1}{2} r (r + 1). \quad [6 \text{ marks}]$$

- (ii) Hence, find the sum

$$S = \sum_{r=1}^n S_r$$

in terms of n . [9 marks]

- (b) The common ratio, r , of a geometric series is given by $r = \frac{5x}{4+x^2}$.

Find all the values of x for which the geometric series converges. [10 marks]

Total 25 marks

4. The function, g , is given by

$$g(x) = 6 - 4x - x^3.$$

- (a) Show that

(i) g is everywhere strictly decreasing [3 marks]

(ii) the equation $6 - 4x - x^3 = 0$ has a real root, β , in the interval $(1, 2)$ [3 marks]

(iii) β is the ONLY real root of the equation. [4 marks]

- (b) Using $x_1 = 2$ as a first approximation to β , illustrate by means of a diagram how the Newton-Raphson method can be used to obtain a second approximation, x_2 , to β . [5 marks]

- (c) Show that if x_n is the n^{th} approximation to β , then

$$x_{n+1} = \frac{2x_n^3 + 6}{3x_n^2 + 4}. \quad [6 \text{ marks}]$$

- (d) Use the Newton-Raphson method to obtain an approximation to β , giving your answer correct to two decimal places. [4 marks]

Total 25 marks

SECTION 3 (MODULE 2.3)

Answer BOTH questions.

5. A recent study of how a representative sample of West Indians travel to work revealed the information shown in the table below.

Type of Transportation	Type of Worker	
	Urban	Rural
Private	400	200
Public	50	20

If a worker is selected at random from this sample, what is the probability that the worker

- (a) is a rural worker? [2 marks]
- (b) uses public transportation? [2 marks]
- (c) uses private transportation? [2 marks]
- (d) is a rural worker or uses public transportation? [5 marks]
- (e) is a rural worker, given he/she uses public transportation? [5 marks]
- (f) uses public transportation, given he/she is an urban worker? [5 marks]
- (g) uses public transportation, given he/she is a rural worker? [4 marks]

Total 25 marks

6. 12,500 copies of a new CD with soca music are sold during the first week of its release in a certain Caribbean island. Sales reached a peak 10 weeks later before dwindling to zero sales in 25 weeks.

- (a) Determine the continuous quadratic function, $S(t)$, to represent weekly sales, t weeks after the release of this CD. [15 marks]
- (b) Use this function to calculate:
 - (i) The total number of CDs sold in the 25-week period [6 marks]
 - (ii) The average number of CDs sold per week [2 marks]
 - (iii) The peak weekly sales [2 marks]

Total 25 marks

END OF TEST